



KOLIN Chemical Co. Ltd.

82, Hwasan 1-gil, Onsan-eup, Ulju-gun, Ulsan, 45009
Republic of Korea

MSDS

CHOLINE CHLORIDE - 75%

§1. Chemical Product and Company Identification

Product Name: Choline chloride - 75%

Chemical Name: Choline chloride

Chemical Family: Quaternary ammonium salt

Chemical Formular: $(\text{CH}_3)_3\text{N}(\text{Cl})\text{CH}_2\text{CH}_2\text{OH}$

Molecular Weight: 139.62

Trade Name and Synonyms:

Choline Chloride 75% Liquid

2-Hydroxyethyl trimethylammonium chloride

Use:

Animal Feed additive, pharmaceuticals;
catalyst, curing agents, neutralizing agent
and miniature printed circuit boards.

Contact Information:

KOLIN Chemical Co., Ltd.

82, Hwasan 1-gil, Onsan-eup, Ulju-gun,

Ulsan, 45009, Republic of Korea

Emergency Telephone Number:

+82-2-584-2191

§2. Composition and Information on Ingredients

Composition:

Name	CAS Number	% by Weight
Choline chloride	67-48-1	More than 75
Water	7732-18-5	Balance

§3. Hazards Identification

Potential Acute Health Effects:

Hazardous in case of ingestion. Hazardous in case of skin contact (irritant), of eye contact (irritant), of inhalation. Slightly hazardous in case of skin contact (permeator).

Potential Chronic Health Effects:

CARCINOGENIC EFFECTS: N/A MUTAGENIC EFFECTS: N/A TERATOGENIC EFFECTS: N/A
DEVELOPMENTAL TOXICITY: N/A The substance is toxic to lungs, mucous membranes. Repeated or prolonged exposure to the substance can produce target organs damage.

§4. First Aid Measures

Eye Contact:

Check for and remove any contact lenses. Immediately flush eyes with running water for at least 15 minutes, keeping eyelids open. Cold water can be used. Do not use an eye ointment. Seek medical attention.

Skin Contact:

After contact with skin, wash immediately with plenty of water. Gently and thoroughly wash the contaminated skin with running water and non-abrasive soap. Be particularly careful to clean folds, crevices, creases and groin. Cold water may be used. Cover the irritated skin with an emollient. If irritation persists, seek medical attention. Wash contaminated clothing before reusing.

Serious Skin Contact:

Wash with a disinfectant soap and cover the contaminated skin with an anti-bacterial cream. Seek medical attention.

Inhalation: Allow the victim to rest in a well ventilated area. Seek immediate medical attention.

Serious Inhalation: N/A

Ingestion:

Do not induce vomiting. Loosen tight clothing such as a collar, tie, belt or wristband. If the victim is not breathing, perform mouth-to-mouth resuscitation. Seek immediate medical attention.

Serious Ingestion: N/A

§5. Fire and Explosion Data

Flammability of the Product: Non-combustible liquid

Auto-Ignition Temperature: N/A

Flash Points: N/A

Flammable Limits: N/A

Product of Combustion:

These products are carbon oxides (CO, CO₂), nitrogen oxides (NO, NO₂...), halogenated compounds.

Fire Hazards in Presence of Various Substances: N/A

Explosion Hazards in Presence of Various Substances:

Risks of explosion of the product in presence of mechanical impact: N/A. Risks of explosion of the product in presence of static discharge: N/A

Fire Fighting Media and Instructions:

SMALL FIRE: Use DRY chemical powder. LARGE FIRE: Use water spray, fog of foam. Do not use water jet.

Special Remarks on Fire Hazards: N/A

Special Remarks on Explosion Hazards: N/A

§6. Accidental Release Measures

Small Spill:

Use appropriate tools to put the spilled solid in a convenient waste disposal container. Finish cleaning by spreading water on the contaminated surface and dispose of according to local and regional authority requirements.

Large Spill:

Use a shovel to put the material into a convenient waste disposal container. Finish cleaning by spreading water on the contaminated surface and allow to evacuate through the sanitary system.

§7. Handling and Storage

Precautions:

Keep away from heat. Keep away from sources of ignition. Ground all equipment containing material. Do not ingest. Wear suitable protective clothing in case of insufficient ventilation, wear suitable respiratory equipment. If ingested, seek medical advice immediately and show the container of the label. Avoid contact with skin and eyes.

Storage:

Keep container dry. Keep in a cool place. Ground all equipment containing material. Keep container tightly closed. Keep in a cool, well-ventilated place. Protect against physical damage.

§8. Exposure Controls / Personal Protection

Engineering Controls:

Use process enclosures, local exhaust ventilation, or other engineering controls to keep airborne levels below recommended exposure limits. If user operations generate dust, fume or mist, use ventilation to keep exposure to airborne contaminants below the exposure limit.

Personal Protection:

Splash goggles. Lab coat. Dust respirator. Be sure to use an approved/certified respirator or equipment. Gloves.

Personal Protection in Case of a Large Spill:

Splash goggles. Full suit. Dust respirator. Boots. Gloves. A self contained breathing apparatus should be used to avoid inhalation of the product. Suggested protective clothing might not be sufficient; consult a specialist BEFORE handling this product.

Exposure Limits: N/A

§9. Physical and Chemical Properties

Physical state and appearance: Liquid

Odor: Slight amine odor

Taste: Not available.

Molecular Weight: 139.62 g/mole

Color: Light amber color

pH (1% soln/water): 6.5 ~ 8.5

Boiling Point: Not available.

Melting Point: Not available.

Critical Temperature: Not available.

Specific Gravity: Not available.

Vapor Pressure: Not applicable.

Vapor Density: Not available.

Volatility: Not available.

Odor Threshold: Not available.

Water/Oil Dist. Coeff.: Not available.

Ionicity (in Water): Not available.

Dispersion Properties: See solubility in water.

Solubility: Soluble in cold water.

§10. Stability and Reactivity Data

Stability: The product is stable.

Instability Temperature: Not available.

Conditions of Instability: Not available.

Corrosivity: Non-corrosive in presence of glass.

Incompatibility with various substances: Not available.

Special Remarks on Reactivity: Not available.

Special Remarks on Corrosivity: Not available.

Polymerization: No.

§11. Toxicological Information

Routes of Entry: Eye contact. Inhalation. Ingestion.

Toxicity to Animals: Acute oral toxicity (LD50): 3400 mg/kg [Rat].

Chronic Effects on Humans: The substance is toxic to lungs, mucous membranes.

Other Toxic Effects on Humans:

Very hazardous in case of ingestion. Hazardous in case of skin contact (irritant), of inhalation.
Slightly hazardous in case of skin contact (permeator).

Special Remarks on Toxicity to Animals: Not available.

Special Remarks on Chronic Effects on Humans: Not available.

Special Remarks on other Toxic Effects on Humans: Not available.

§12. Ecological Information

Ecotoxicity: Not available.

BOD and COD: Not available.

Products of Biodegradation:

Possibly hazardous short term degradation products are not likely. However, long term degradation products may arise.

Toxicity of the Products of Biodegradation: The products of degradation are more toxic.

Special Remarks on the Products of Biodegradation: Not available.

§13. Disposal Considerations

Waste Disposal

Incinerate in a chemical incinerator where permitted under Federal, Provincial and local regulations.

§14. Transport Information

DOT Classification: Not a DOT controlled material (United States).

Identification: Not applicable.

Special Provisions for Transport: Not applicable.

This product is NOT classified as dangerous for IATA Transport.

§15. Other Regulatory Information

Canada (NDSL)	Not Determined
China (IECSC)	Not Determined
Europe (EINECS)	Not Determined
Europe (REACH)	Not Determined
Japan (ENCS/METI)	Not Determined
Korea (KECI)	Not Determined
Malaysia (EHS Register)	Not Determined
Philippines (PICCS)	Not Determined
Taiwan (NCSR)	Not Determined
USA (TSCA)	Not Determined

§16. Other Information

References: Not available.

Other Special Considerations: Not available.

Created: January, 2010

Last Updated: June, 2016

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